UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	Authoring tools help developers organise and edit multimedia elements.	Authoring tools assemble media.
A2	FALSE	3D animation does not use modelling.	3D animation relies on models.
A3	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A4	TRUE	Hertz measures sound frequency.	Hertz is cycles per second.
A5	FALSE	A vector image becomes pixelated when enlarged.	Vector images scale without pixelation.
A6	TRUE	MPEG is a common family of video compression formats.	MPEG defines video compression standards.
A7	FALSE	Anti-aliasing makes the edges of text look more jagged.	Anti-aliasing smooths jagged edges.
A8	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A9	FALSE	MIDI files store digitized audio samples.	MIDI stores musical instructions, not samples.
A10	FALSE	A plotter is an input device.	A plotter produces output.
A11	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.
A12	TRUE	A web browser displays hypertext documents.	Browsers render web pages.
A13	TRUE	USB is a plug-and-play connection standard.	USB devices are recognised automatically.
A14	TRUE	RGB is an additive colour model.	RGB adds red, green and blue light.
A15	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	J	Multimedia	integration of text, graphics, audio, video and animation
B2	M	Virtual reality	simulated environment
B3	A	Frame rate	frames displayed per second
B4	I	Bitmap	image made of pixels
B5	O	Storyboard	visual sequence plan
B6	L	Keyframe	important frame in an animation
B7	C	Browser	software that displays web pages
B8	G	Scanner	input device that digitises documents
B9	D	Linear multimedia	presentation without user control of sequence
B10	F	Vector	image made of mathematical objects
B11	E	Hypermedia	linked multimedia elements
B12	B	HTML	language for web structure
B13	H	Copyright	exclusive rights of a creator
B14	N	Tweening	generating in-between frames
B15	K	Card-based tool	authoring tool organised like pages or cards

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	A low-fidelity layout sketch
C2	A	Card-based
C3	B	Links to other content
C4	C	WAV
C5	C	Compiler
C6	C	Structure web content
C7	B	Text, graphics, audio, video and animation
C8	A	One shape into another
C9	B	Temporary working memory
C10	C	Frames per second
C11	B	PNG
C12	A	In-between frames
C13	B	An early model of a product
C14	A	Draws alternate lines of a frame
C15	D	It combines visuals and sound
C16	C	Screen display
C17	A	Scanner
C18	A	Non-linear
C19	D	Analog sound to digital values
C20	B	Modelling and rendering
C21	B	Small strokes at character ends
C22	C	Important frames in an animation
C23	B	Messages with a beginning and end
C24	B	CMYK
C25	C	Frequency
C26	C	A cinema presentation
C27	B	Loudness
C28	A	Another resource
C29	A	Merging of media and computing
C30	D	Automatic device recognition

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PART D - WORKED ANSWERS

QUESTION 1

Topic: text

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

Sans serif is cleaner at screen sizes; serif aids reading in print.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

It links related content and supports non-linear navigation.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

(4 marks)

QUESTION 2

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(5 marks)

QUESTION 3

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

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QUESTION 4

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

END OF MARKING SCHEME

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